Empirical Results & Trade-Off Analysis: FDA-LoRA Toy Model Benchmarks

Evaluation of NEW2FDA and NEW3FDA against Federated Baselines

FDA-LoRA Benchmark Analysis Suite

August 2026

1 Experimental Configuration & Scope

This report presents numerical results, convergence metrics, compute profiles, and empirical findings from the toy benchmark evaluation suite.

1.1 Run Count Accounting

The evaluation comprises exactly **66 complete runs** across 3 random seeds ($S \in \{42, 43, 44\}$), each executed for $T = 100$ communication rounds:

$$\underbrace{2 \text{ non-linear scenarios} \times 6 \text{ methods} \times 3 \text{ seeds}}_{36 \text{ runs}} + \underbrace{2 \text{ linear scenarios} \times 5 \text{ methods} \times 3 \text{ seeds}}_{30 \text{ runs}} = 66 \text{ runs}. \quad (1)$$

The FDA SMA variant was evaluated exclusively in the non-linear scenarios (`NEW3FDA`), whereas all other methods were evaluated across all four scenarios. All deprecated exploratory sweeps and duplicate legacy runs are strictly excluded.

1.2 Benchmark Specifications

- **Dataset**: MNIST (60,000 training examples, 10,000 test examples, 10 classes).
- **Architectures**:
  1. LIN: Single-layer linear projection ($\mathbb{R}^{784} \to \mathbb{R}^{10}$).
  2. NONLIN: Two-layer MLP ($784 \to 128 \to 10$, ReLU activation).
- **Data Partitions**:
  1. `10c1`: 10 clients, 1 class per client (pathological non-IID; zero class overlap).
  2. `5c2`: 5 clients, 2 classes per client (partitioned non-IID).
- **Optimization & LoRA Hyperparameters**: Linear runs used SGD with $\eta = 10^{-2}$; non-linear runs used AdamW with $\eta = 10^{-3}$ (weight decay $\lambda = 10^{-4}$). LoRA configuration: rank $r = 4$, scaling $\alpha = 8$, batch size $B = 64$.
- **Evaluated Methods**:
  - **LoRA** (`FedAvg-LoRA`): Static local computation budget ($E = 5$ epochs/round).
  - **FFA-LoRA**: Static local computation ($E = 5$), matrix A frozen after initialization, updating only matrix B.
  - **RoLoRA** (`FedAvg-RoLoRA`): Alternating optimization and aggregation of LoRA matrices A and B, with the associated orthogonalization step ($E = 5$).
  - **FDA Standard (`NEW2FDA`)**: Adaptive epoch schedule $E_t \in [E_{\min}, E_{\max}] = [3, 15]$ driven by instantaneous drift variance.
  - **FDA EMA-Standard (`NEW2FDA`)**: Adaptive epoch schedule with Exponential Moving Average ($\rho = 0.9$) smoothed variance.

  - **FDA SMA (`NEW3FDA`)**: Adaptive epoch schedule with Simple Moving Average ($W = 3$ rounds) smoothed variance.

- **Benchmark Adaptation Notice**: This experimental setup reproduces the data-partitioning and alternating-training concepts while adapting the network architecture ($784 \to 128 \to 10$ MLP) and LoRA configuration ($r = 4$) rather than serving as an exact replication of the original RoLoRA toy setup (which utilized $r = 16$ and a fixed W_{out} layer).

2 Empirical Accuracy Results

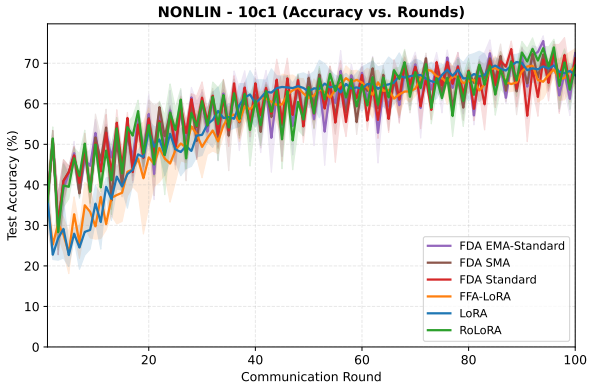
Table 1: Fixed-round test accuracy (%, Mean ± Std across 3 seeds) at communication rounds 25, 50, 75, and 100.

Architecture	Method	Round 25	Round 50	Round 75	Round 100
Nonlin - `10c1`	**FDA EMA-Standard**	50.64 ± 2.57	65.17 ± 2.78	61.99 ± 4.25	$\mathbf{72.47 \pm 0.85}$
	FDA Standard	49.29 ± 4.41	63.95 ± 2.94	64.52 ± 1.58	71.23 ± 0.41
	FDA SMA ($W = 3$)	48.64 ± 4.80	$\mathbf{66.41 \pm 1.68}$	62.68 ± 1.38	69.43 ± 2.88
	RoLoRA	$\mathbf{53.51 \pm 5.45}$	64.13 ± 1.77	61.38 ± 0.61	69.40 ± 3.58
	FFA-LoRA	48.54 ± 7.91	61.43 ± 1.37	66.39 ± 2.40	69.14 ± 4.42
	LoRA	48.77 ± 2.94	63.03 ± 4.02	$\mathbf{67.11 \pm 2.63}$	67.01 ± 2.07
Nonlin - `5c2`	**FDA Standard**	$\mathbf{66.05 \pm 8.94}$	72.26 ± 1.14	72.70 ± 4.45	$\mathbf{76.89 \pm 4.65}$
	RoLoRA	65.62 ± 2.22	$\mathbf{75.69 \pm 2.67}$	72.42 ± 4.45	75.73 ± 5.39
	LoRA	65.15 ± 1.73	68.93 ± 2.56	67.17 ± 2.11	75.07 ± 1.05
	FDA SMA ($W = 3$)	65.62 ± 3.05	75.22 ± 2.21	$\mathbf{73.82 \pm 2.42}$	72.80 ± 6.62
	FDA EMA-Standard	61.95 ± 2.82	69.65 ± 1.13	61.34 ± 1.51	71.41 ± 3.22
	FFA-LoRA	57.02 ± 0.17	57.61 ± 0.96	58.28 ± 1.60	59.22 ± 2.03
Lin - `10c1`	**LoRA**	$\mathbf{56.35 \pm 9.54}$	$\mathbf{62.13 \pm 14.47}$	$\mathbf{63.13 \pm 15.43}$	$\mathbf{68.01 \pm 10.76}$
	FDA Standard	40.00 ± 10.96	50.04 ± 5.11	44.62 ± 11.53	55.77 ± 5.52
	RoLoRA	40.46 ± 1.26	46.89 ± 3.58	49.14 ± 8.71	54.83 ± 7.05
	FDA EMA-Standard	40.01 ± 7.77	50.59 ± 7.25	47.36 ± 6.05	54.65 ± 2.06
	FFA-LoRA	27.08 ± 5.02	25.64 ± 7.97	29.74 ± 13.10	31.17 ± 7.52
Lin - `5c2`	**RoLoRA**	42.36 ± 0.35	$\mathbf{55.30 \pm 2.04}$	48.18 ± 12.91	$\mathbf{61.26 \pm 3.55}$
	FDA Standard	$\mathbf{45.76 \pm 2.89}$	55.20 ± 5.44	$\mathbf{52.08 \pm 3.08}$	57.87 ± 3.22
	FDA EMA-Standard	37.10 ± 3.38	51.44 ± 1.92	50.08 ± 3.67	55.08 ± 5.18
	LoRA	42.69 ± 3.97	46.89 ± 6.01	43.10 ± 2.89	43.94 ± 1.77
	FFA-LoRA	32.65 ± 2.02	31.44 ± 2.35	33.16 ± 4.40	31.55 ± 5.34

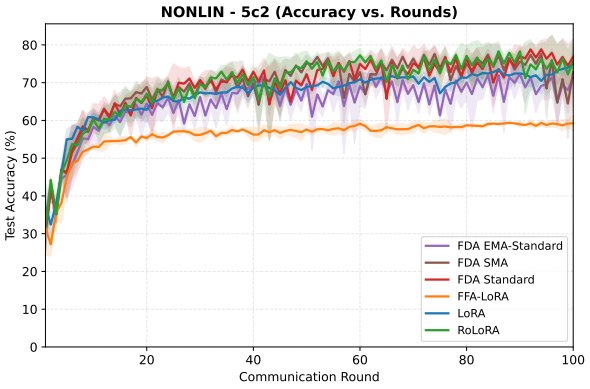
Table 2: Peak test accuracy and best communication round distribution across 100 rounds.

Architecture	Method	Peak Accuracy (%)	Mean Best Round	Median Best Round	Min / Max Best Round
Nonlin - 10c1	**FDA EMA-Standard**	$\mathbf{75.49 \pm 1.30}$	94.0	94	[94, 94]
	RoLoRA	75.24 ± 0.44	93.3	92	[92, 96]
	FDA Standard	74.56 ± 1.24	94.7	98	[88, 98]
	FDA SMA ($W = 3$)	73.56 ± 1.51	88.7	86	[84, 96]
	LoRA	71.82 ± 1.49	85.7	85	[82, 90]
	FFA-LoRA	71.78 ± 1.22	96.7	98	[92, 100]
Nonlin - 5c2	**FDA Standard**	$\mathbf{80.07 \pm 1.90}$	93.3	94	[90, 96]
	RoLoRA	79.93 ± 0.34	93.3	96	[86, 98]
	FDA SMA ($W = 3$)	79.33 ± 0.98	77.3	74	[74, 84]
	LoRA	75.12 ± 1.06	97.0	100	[91, 100]
	FDA EMA-Standard	74.99 ± 1.61	76.0	82	[56, 90]
	FFA-LoRA	60.14 ± 0.69	93.0	92	[87, 100]
Lin - 10c1	**LoRA**	$\mathbf{70.98 \pm 9.60}$	91.7	97	[80, 98]
	RoLoRA	60.12 ± 1.82	88.0	88	[76, 100]
	FDA EMA-Standard	59.72 ± 1.60	84.3	85	[82, 86]
	FDA Standard	59.47 ± 1.36	80.7	72	[70, 100]
	FFA-LoRA	39.12 ± 4.40	51.0	75	[1, 77]
Lin - 5c2	**RoLoRA**	$\mathbf{65.91 \pm 0.77}$	75.3	80	[62, 84]
	FDA Standard	64.80 ± 2.20	90.3	91	[88, 92]
	FDA EMA-Standard	62.52 ± 1.99	84.7	94	[66, 94]
	LoRA	54.48 ± 1.84	45.3	35	[33, 68]
	FFA-LoRA	38.24 ± 2.51	49.0	40	[21, 86]

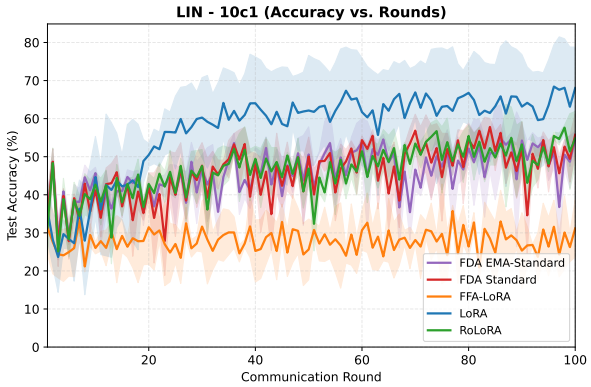
Methodological Warning on Peak Accuracy: Selecting the peak communication round based on test-set accuracy introduces test-set selection bias (post-hoc checkpoint snooping). Peak accuracy metrics reported in Table 2 are purely descriptive. Formal comparative claims must rely on fixed-round evaluations (Table 1, such as Round 100) or checkpoints chosen via a client-side validation split.



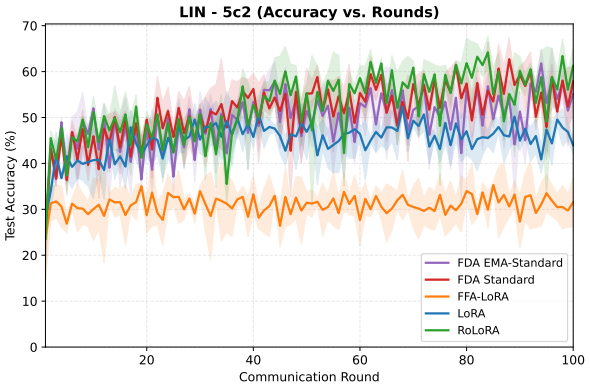
(a) Nonlin - `10c1` (10 clients, 1 class/client)



(b) Nonlin - `5c2` (5 clients, 2 classes/client)



(c) Lin - `10c1` (10 clients, 1 class/client)



(d) Lin - `5c2` (5 clients, 2 classes/client)

Figure 1: Complete test accuracy trajectories across 100 communication rounds for all four benchmark configurations.

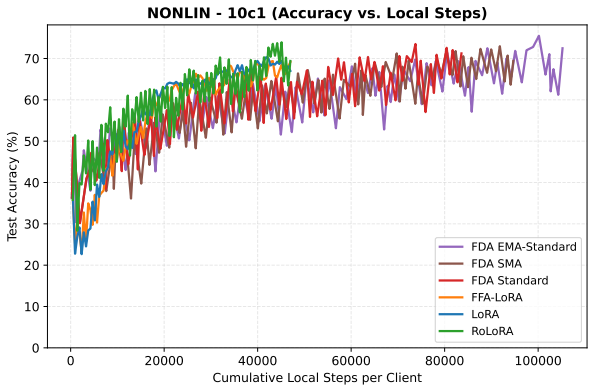
3 Communication vs. Computation Trade-Offs

Table 3: Communication rounds required to reach target test accuracy milestones (Mean ± Std round across successful seeds; success ratio $N_{\text{success}}/N_{\text{total}}$).

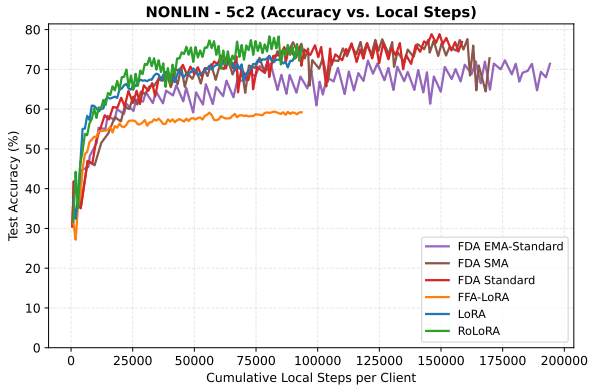
Scenario	Method	Target 60.0%	Target 65.0%	Target 70.0%	Target 75.0%	Target 77.5%
Nonlin - 10c1	**FDA EMA-Standard**	26.0 ± 4.3 (3/3)	43.3 ± 7.7 (3/3)	65.3 ± 13.2 (3/3)	89.0 ± 5.0 (2/3)	– (0/3)
	FDA Standard	26.7 ± 0.9 (3/3)	38.0 ± 7.5 (3/3)	$\mathbf{56.7 \pm 10.6}$ (3/3)	98.0 ± 0.0 (1/3)	– (0/3)
	FDA SMA ($W = 3$)	22.7 ± 2.5 (3/3)	42.0 ± 6.5 (3/3)	64.7 ± 3.8 (3/3)	– (0/3)	– (0/3)
	RoLoRA	$\mathbf{21.3 \pm 2.5}$ (3/3)	40.7 ± 7.4 (3/3)	67.3 ± 4.1 (3/3)	94.0 ± 2.0 (2/3)	– (0/3)
	LoRA	35.7 ± 4.5 (3/3)	44.3 ± 4.1 (3/3)	71.0 ± 13.4 (3/3)	– (0/3)	– (0/3)
	FFA-LoRA	39.3 ± 1.7 (3/3)	49.0 ± 3.3 (3/3)	91.3 ± 4.9 (3/3)	– (0/3)	– (0/3)
Nonlin - 5c2	**FDA SMA** ($W = 3$)	$\mathbf{10.0 \pm 1.6}$ (3/3)	$\mathbf{15.7 \pm 1.7}$ (3/3)	$\mathbf{23.3 \pm 2.5}$ (3/3)	$\mathbf{42.3 \pm 5.4}$ (3/3)	62.7 ± 9.8 (3/3)
	FDA Standard	10.7 ± 2.5 (3/3)	17.7 ± 4.5 (3/3)	26.7 ± 3.4 (3/3)	52.7 ± 8.2 (3/3)	69.3 ± 14.8 (3/3)
	RoLoRA	13.0 ± 4.1 (3/3)	17.7 ± 3.3 (3/3)	29.3 ± 5.2 (3/3)	45.7 ± 9.7 (3/3)	$\mathbf{55.7 \pm 3.3}$ (3/3)
	FDA EMA-Standard	12.7 ± 3.4 (3/3)	23.3 ± 7.7 (3/3)	37.7 ± 2.9 (3/3)	84.0 ± 6.0 (2/3)	– (0/3)
	LoRA	9.7 ± 0.9 (3/3)	20.3 ± 3.1 (3/3)	43.3 ± 6.6 (3/3)	93.0 ± 7.0 (2/3)	– (0/3)
	FFA-LoRA	78.5 ± 18.5 (2/3)	– (0/3)	– (0/3)	– (0/3)	– (0/3)

Table 4: Cumulative computation steps and local epoch allocation over 100 communication rounds (step counts denote synchronized local gradient steps per client).

Scenario	Method	Cumulative local steps / client	Local steps / client to peak round	Mean Local Epochs
Nonlin - 10c1	**FDA EMA-Standard**	105,217	100,173	11.19
	FDA SMA ($W = 3$)	94,689	84,976	10.07
	FDA Standard	83,597	79,242	8.89
	RoLoRA	47,000	43,867	5.00
	LoRA	47,000	40,263	5.00
	FFA-LoRA	47,000	45,433	5.00
Nonlin - 5c2	**FDA EMA-Standard**	194,168	150,722	10.38
	FDA SMA ($W = 3$)	169,671	132,334	9.07
	FDA Standard	158,514	148,166	8.48
	RoLoRA	93,500	87,267	5.00
	LoRA	93,500	90,695	5.00
	FFA-LoRA	93,500	86,955	5.00



(a) NONLIN - `10c1` (Accuracy vs. Local Steps per Client)



(b) NONLIN - `5c2` (Accuracy vs. Local Steps per Client)

Figure 2: Test accuracy trajectories as a function of cumulative local gradient steps per client.

4 FDA Variance Dynamics & Epoch Boundary Distribution

Definition of Mean Variance: The parameter drift variance metric $\bar{V}$ represents the empirical variance of local client adapter updates relative to the global model, averaged across all participating clients, communication rounds $t \in [1, 100]$, random seeds ($S \in \{42, 43, 44\}$), and LoRA weight tracks (matrices A and B).

Table 5: FDA internal drift variance dynamics and epoch boundary allocations ($E_{\min} = 3, E_{\max} = 15$).

Scenario	FDA Variant	Mean Variance	Mean Epochs	Median Epochs	Std Epochs	Frac $E_{\min}$ (3)	Frac $E_{\max}$ (15)
Nonlin - `10c1`	**FDA EMA-Standard**	5.13±4.42	11.19	12.5	4.26	11.7%	44.3%
	FDA SMA ($W = 3$)	5.05±4.86	10.07	10.0	4.62	16.0%	38.7%
	FDA Standard	4.98±4.81	8.89	8.0	5.29	29.7%	37.3%
Nonlin - `5c2`	**FDA EMA-Standard**	5.03±4.29	10.38	10.0	4.37	10.0%	41.0%
	FDA SMA ($W = 3$)	4.77±5.02	9.07	9.0	3.55	9.7%	16.0%
	FDA Standard	4.67±4.75	8.48	8.0	3.85	17.0%	17.3%
Lin - `10c1`	**FDA EMA-Standard**	6.95±5.82	11.23	15.0	4.77	14.7%	54.3%
	FDA Standard	6.94±6.28	9.05	7.0	5.91	43.3%	49.3%
Lin - `5c2`	**FDA EMA-Standard**	26.77 ± 19.94	10.62	15.0	5.08	21.0%	50.7%
	FDA Standard	25.99 ± 20.83	8.99	8.0	5.75	41.3%	45.3%

5 Empirical Conclusions & Discussion

1. **Non-Linear Benchmark Performance**:
   - On NONLIN-`10c1`, **FDA EMA-Standard** yields the numerically highest final accuracy (72.47% ± 0.85%) at Round 100, compared to LoRA (67.01% ± 2.07%), FFA-LoRA (69.14% ± 4.42%), and RoLoRA (69.40% ± 3.58%). However, **FDA Standard** exhibits lower cross-seed standard deviation (±0.41%) with a final accuracy of 71.23% ± 0.41%.
   - On NONLIN-`5c2`, **FDA Standard** reaches the numerically highest peak accuracy (80.07%±1.90%) and final accuracy (76.89% ± 4.65%), closely followed by **RoLoRA** (79.93% ± 0.34% peak, 75.73% ± 5.39% final). Across 3 seeds, this 0.14 percentage-point peak difference is statistically indistinguishable.
2. **Milestone Convergence & Success Rate Qualification**:

- For the 75.0% target milestone on NONLIN-5c2, **FDA SMA** achieves the threshold in a conditional mean of 42.3 ± 5.4 rounds across all 3/3 seeds, whereas **LoRA** averages 93.0 ± 7.0 rounds among the 2/3 seeds that reached it (1/3 seed failed to achieve 75%).
- The apparent 2.20× speedup represents a comparison over successful runs and must be interpreted alongside the higher success ratio (3/3 vs 2/3).

3. **Variance Smoothing Dynamics**:
   - EMA filtering ($\rho = 0.9$) smooths epoch allocation transitions (11.7% at $E_{\min}$, 44.3% at $E_{\max}$) relative to raw FDA Standard (29.7% at $E_{\min}$, 37.3% at $E_{\max}$).
   - SMA filtering ($W = 3$) maintains stable local computation (9.07 mean epochs on 5c2) while avoiding extreme boundary saturation (9.7% at $E_{\min}$, 16.0% at $E_{\max}$).
4. **Linear Regime Observations**:
   - On LIN-10c1, **LoRA** attains a final accuracy of $68.01\% \pm 10.76\%$ with high cross-seed variability, while **FDA Standard** and **RoLoRA** finish at Round 100 with 55.77%±5.52% and 54.83%±7.05%.
   - On LIN-5c2, **RoLoRA** ($61.26\% \pm 3.55\%$) and **FDA Standard** ($57.87\% \pm 3.22\%$) achieve higher final accuracy than **LoRA** ($43.94\% \pm 1.77\%$).
5. **Compute vs. Communication Trade-Off Assessment**:
   - FDA executes an average of 8.5 to 11.2 local epochs per round (1.7× to 2.2× baseline compute), expending 83.6k–105.2k cumulative local steps per client on 10c1 and 158.5k–194.2k local steps per client on 5c2 vs baseline fixed-epoch (47.0k and 93.5k local steps per client).
   - FDA may be advantageous when communication latency or bandwidth is substantially more costly than additional local computation; this benchmark did not directly measure network cost.